LECTURE 15 · DISCRETE MATHEMATICS

Repair relations. Discover classes.

Closures, modular arithmetic, equivalence relations, and partitions

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A guided set of course notes

We begin by repairing a relation with the smallest possible additions. That idea leads to reachability, modular arithmetic, and finally the way an equivalence relation divides a set into nonoverlapping classes.

How to read these notes. Slide-derived material is presented as the main narrative. Any new context, extension, or extra example is labeled **[Added]**. Any unresolved gap in the slide material is labeled **[Unclear]**.

Based on Lecture 15 and Rosen, 7th ed., §§9.4, 4.1, 4.4, and 9.5

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Learning goals

After working through these notes, you should be able to:

- compute reflexive, symmetric, and transitive closures and justify why no unnecessary pairs were added;
- interpret the transitive closure as reachability by positive-length paths;
- calculate with congruences and explain why modular division needs an inverse;
- prove that a relation is, or is not, an equivalence relation;
- find equivalence classes and move in both directions between equivalence relations and partitions; and
- solve the lecture's challenge problems using the same core ideas.

The route through the lecture

The lecture has one unifying question: *what structure appears when we enforce rules on a relation?* First, a closure enforces one rule with the fewest new pairs. Congruence modulo m then supplies a rich example of a relation. Finally, reflexivity, symmetry, and transitivity together make every element belong to exactly one equivalence class.

1 Closures: the smallest complete repair

Slide sequence: Slides 1–13

Textbook anchor: §9.4, "Closures of Relations," pp. 597–601.

Let R be a relation on a set A. A **closure of R with respect to a property** P is a relation C satisfying three requirements:

1. **Containment:** $R \subseteq C$. Every original pair survives.
2. **Correctness:** C has property P.
3. **Minimality:** if S contains R and has property P, then $C \subseteq S$.

The third condition is easy to overlook. A large relation such as $A \times A$ is reflexive, symmetric, and transitive, but it is usually not *the* requested closure. "Smallest" means smallest by set inclusion, not merely a relation with few pairs by accident.

1.1 Reflexive closure: supply every missing loop

A relation on A is **reflexive** when $(a, a) \in R$ for every $a \in A$. The **diagonal relation** on A is

$$\Delta_A = \{(a, a) : a \in A\}.$$

Consequently, the reflexive closure is

$$r(R) = R \cup \Delta_A.$$

This formula automatically preserves R, adds every required loop, and adds nothing else. Any reflexive relation containing R must contain all of Δ_A, so it must contain $R \cup \Delta_A$.

Worked example: repair the lecture's opening relation

Let

$$A = \{a, b, c\}, \qquad R = \{(a, a), (a, b), (b, a), (b, c)\}.$$

Inspect the diagonal pairs $(a, a), (b, b), (c, c)$. The first is already present; the other two are missing. Therefore

$$r(R) = R \cup \{(b, b), (c, c)\}.$$

In the directed-graph picture, (a, a) is already a loop at a. The repair adds a loop at b and a loop at c. No reverse arrows or path shortcuts are needed because reflexivity asks only about loops.

[Added] Worked example: a relation given by an inequality

On $\mathbb{Z}$, define aRb when $a < b$. This relation is not reflexive because $a < a$ is false. Adding exactly the pairs (a, a) changes the condition from $a < b$ to $a \leq b$. Thus the reflexive closure is

$$\{(a, b) \in \mathbb{Z}^2 : a \leq b\}.$$

This is adapted from Rosen §9.4, Example 1, p. 598.

1.2 Symmetric closure: supply every reverse arrow

A relation is **symmetric** when

$$(a, b) \in R \Longrightarrow (b, a) \in R.$$

The **inverse relation** is

$$R^{-1} = \{(b, a) : (a, b) \in R\}.$$

Therefore the symmetric closure is

$$s(R) = R \cup R^{-1}.$$

Loops cause no special difficulty: the reverse of (a, a) is the same pair. Similarly, if both directions are already present, neither needs to be added.

Worked example: reverse only the unsupported arrow

For the opening relation, (a, b) and (b, a) already support each other, and (a, a) supports itself. The pair (b, c) has no reverse. Hence

$$s(R) = R \cup \{(c, b)\}.$$

The graph gains one arrow, from c back to b.

[Added] Worked example: strict comparison loses its direction

On the positive integers, let aRb mean $a > b$. The inverse relation contains exactly the

pairs with $a < b$. Their union contains every off-diagonal pair:

$$s(R) = \{(a, b) : a \neq b\}.$$

It does not contain loops because symmetry never requires (a, a). This is adapted from Rosen §9.4, Example 2, p. 598.

1.3 Transitive closure: turn every route into a shortcut

A relation is **transitive** when

$$(x, y) \in R \text{ and } (y, z) \in R \Longrightarrow (x, z) \in R.$$

The **transitive closure**, written $t(R)$ or R^*, is the smallest transitive relation containing R.

The most useful interpretation is reachability:

$$(x, y) \in t(R) \quad \Longleftrightarrow \quad \text{there is an } R\text{-path of length at least one from } x \text{ to } y.$$

An original edge is a path of length one. Two paths can be concatenated, which explains transitivity. Finally, every transitive relation containing R must contain a shortcut for every path, which explains minimality.

The phrase *length at least one* matters. A loop (x, x) enters $t(R)$ only when there is a positive-length route from x back to itself. The empty path is not used for this closure.

Worked example: finish the opening repair

Start again with

$$R = \{(a, a), (a, b), (b, a), (b, c)\}.$$

There is a path $a \to b \to c$, so (a, c) is required. There is also a path $b \to a \to b$, so (b, b) is required. After adding these pairs,

$$t(R) = R \cup \{(a, c), (b, b)\}.$$

To check completeness, list reachable endpoints:

- from a, the vertices a, b, c are reachable;
- from b, the vertices a, b, c are reachable;
- from c, no positive-length path begins.

All corresponding pairs are already present, so the repair is complete.

Exam lens: do not stop after one round

Adding shortcuts can create longer routes whose shortcuts were not visible in the original relation. A reliable method is to compute all reachable endpoints or to repeat the repair until an entire pass adds nothing.

Worked example: a chain needs a length-three shortcut

For

$$R = \{(1,2),(2,3),(3,4)\},$$

paths of length two force $(1,3)$ and $(2,4)$. The length-three path $1 \to 2 \to 3 \to 4$ also forces $(1,4)$. Therefore

$$t(R) = R \cup \{(1,3),(2,4),(1,4)\}.$$

Adding only the length-two shortcuts would miss $(1,4)$.

[Added] Worked example: a practical reachability interpretation

Suppose $(u,v) \in R$ means there is a direct one-way data link from server u to server v. Then $(u,v) \in R^2$ means a message can travel using two direct links, and $(u,v) \in R^*$ means it can travel along some positive number of links. The transitive closure is therefore the all-pairs reachability table. This interpretation follows Rosen §9.4, pp. 600–601.

1.4 When symmetry and transitivity create loops

Symmetry plus transitivity can force a loop, but only at a vertex touching an edge. If cRd, symmetry gives dRc, and transitivity applied to cRd and dRc gives cRc.

The slide condition

$$\forall a \in A\ \exists b \in A : aRb \text{ or } bRa$$

says that every vertex is incident with at least one edge. After symmetric and then transitive closure, each vertex lies on a two-edge return route. Hence $t(s(R))$ is reflexive on A.

The condition is essential. For $A = \{a,b,c\}$ and $R = \{(a,b)\}$,

$$t(s(R)) = \{(a,a),(a,b),(b,a),(b,b)\}.$$

The vertex c is isolated, so (c,c) is not forced.

1.5 A compact comparison of the three repairs

Closure	What must be added?	What may remain absent?
Reflexive	Every missing loop (a,a)	Reverse arrows and shortcuts
Symmetric	Every missing reverse (b,a)	Loops at isolated vertices
Transitive	Every shortcut for a positive-length path	Loops without a return path

Always state the underlying set. Reflexive closure depends on all elements of that set, including elements not appearing in any original pair.

2 Modular arithmetic: arithmetic that cycles

Slide sequence: Slides 14–19

Textbook anchor: §4.1, pp. 240–242; §4.3, p. 272; §4.4, pp. 275–276.

Fix a positive integer m, called the **modulus**. Integers a and b are **congruent modulo** m when m divides their difference:

$$a \equiv b \pmod{m} \iff m \mid (a-b).$$

Equivalently, a and b leave the same remainder after division by m.

For example,

$$12 \equiv 2 \pmod{5} \qquad \text{because} \qquad 12 - 2 = 10 = 2 \cdot 5.$$

Negative integers cause no difficulty. The standard remainder lies in $\{0, 1, \ldots, m-1\}$, so

$$-3 = (-1) \cdot 5 + 2 \qquad \Longrightarrow \qquad -3 \equiv 2 \pmod{5}.$$

[Added] Congruence and remainder are related but not identical

The statement $a \equiv b \pmod{m}$ is a relation between two integers. The value $a \bmod m$ is a function value: the unique remainder in $\{0, \ldots, m-1\}$. They connect through

$$a \equiv b \pmod{m} \iff (a \bmod m) = (b \bmod m).$$

This distinction is emphasized in Rosen §4.1, pp. 240–241.

2.1 Safe operations: add, subtract, multiply, then reduce

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m}, \qquad ac \equiv bd \pmod{m}.$$

Subtraction follows from addition by using $-c \equiv -d \pmod{m}$. This means we may replace large operands by their smaller representatives before computing.

Worked example: the three calculations from the lecture

Work modulo 5 and choose answers in $\{0, 1, 2, 3, 4\}$.

$$\begin{aligned} 18 + 14 &\equiv 3 + 4 = 7 \equiv 2 \pmod{5}, \\ 2 - 9 &\equiv 2 - 4 = -2 \equiv 3 \pmod{5}, \\ 13 \cdot 12 &\equiv 3 \cdot 2 = 6 \equiv 1 \pmod{5}. \end{aligned}$$

In the second line, adding 5 to -2 gives the standard representative 3.

[Added] Worked example: a clock calculation

On a 24-hour clock, suppose the current time is 19:00. What is the time 53 hours later?

$$19 + 53 = 72 \equiv 0 \pmod{24}.$$

Therefore the clock reads 00:00. The modulus models the fact that the display returns to the same state every 24 hours.

2.2 Division is multiplication by an inverse

An integer u is a **multiplicative inverse of** b **modulo** m when

$$bu \equiv 1 \pmod{m}.$$

For $m > 1$, such an inverse exists exactly when $\gcd(b, m) = 1$. If it exists, we may multiply both sides of a congruence by it.

Worked example: solve a linear congruence by an inverse

Solve $3x \equiv 12 \pmod{7}$. Since

$$3 \cdot 5 = 15 \equiv 1 \pmod{7},$$

the inverse of 3 modulo 7 is 5. Multiply both sides by 5:

$$x \equiv 5 \cdot 12 = 60 \equiv 4 \pmod{7}.$$

Check: $3 \cdot 4 = 12$, so the result satisfies the original congruence.

Worked example: why ordinary cancellation can lose answers

Solve $2x \equiv 2 \pmod{6}$ for $0 \le x < 6$. Testing the six residues gives

$$2(1) = 2 \equiv 2 \pmod{6}, \qquad 2(4) = 8 \equiv 2 \pmod{6}.$$

Thus $x = 1$ and $x = 4$ both work. The number 2 has no inverse modulo 6 because $\gcd(2, 6) = 2 \neq 1$. Cancelling 2 as if this were an ordinary equation would produce only $x \equiv 1 \pmod{6}$ and incorrectly discard $x = 4$.

[Added] Worked example: find an inverse with Bézout's identity

Find an inverse of 17 modulo 43, then solve $17x \equiv 9 \pmod{43}$.
The Euclidean algorithm gives

$$43 = 2 \cdot 17 + 9, quad 17 = 1 \cdot 9 + 8, quad 9 = 1 \cdot 8 + 1.$$

Work backward:

$$\begin{aligned} 1 &= 9 - 8 \\ &= 9 - (17 - 9) = 2 \cdot 9 - 17 \\ &= 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17. \end{aligned}$$

Therefore $-5 \equiv 38 \pmod{43}$ is an inverse of 17. Multiplying the original congruence by 38 gives

$$x \equiv 38 \cdot 9 = 342 \equiv 41 \pmod{43}.$$

Indeed, $17 \cdot 41 = 697 = 16 \cdot 43 + 9$. This method is developed in Rosen §4.4, pp. 275–276.

3 Equivalence relations: sameness for a chosen purpose

Slide sequence: Slides 20–25

Textbook anchor: §9.5, "Equivalence Relations" and "Equivalence Classes," pp. 608–611.

An **equivalence relation** on a set A is a relation that is reflexive, symmetric, and transitive. Each property contributes part of the meaning of "equivalent":

- reflexivity says every object is equivalent to itself;
- symmetry says equivalence does not have a direction; and
- transitivity says equivalence is consistent through intermediates.

3.1 Why congruence is an equivalence relation

On $\mathbb{Z}$, define aRb by $a \equiv b \pmod{m}$, where $m > 1$.

- **Reflexive:** $a - a = 0 = m \cdot 0$, so $m \mid (a - a)$.
- **Symmetric:** if $a - b = mk$, then $b - a = m(-k)$.
- **Transitive:** if $a - b = mk$ and $b - c = m\ell$, then

$$a - c = (a - b) + (b - c) = m(k + \ell).$$

Thus congruence modulo m is an equivalence relation.

3.2 Equivalence classes

For an equivalence relation R on A, the **equivalence class of** a is

$$[a] = \{x \in A : xRa\}.$$

The element a is a **representative** of the class, but it is not a special member. Every member can name the same class.

Modulo 5,

$$[2] = \{\ldots, -8, -3, 2, 7, 12, \ldots\} = \{5k + 2 : k \in \mathbb{Z}\}.$$

Because $7 \equiv 2 \pmod{5}$ and $-3 \equiv 2 \pmod{5}$, $[2] = [7] = [-3]$.

The five congruence classes modulo 5 are

$$[r] = \{5k + r : k \in \mathbb{Z}\}, \qquad r = 0, 1, 2, 3, 4.$$

Every integer belongs to exactly one of them.

Worked example: compare class names rather than representatives

Among $[-8], [2], [7], [12], [3]$ modulo 5, the first four are equal because their representatives all leave remainder 2. The class $[3]$ is different. Therefore

$$[2] \cap [3] = \varnothing, \qquad [2] \cup [7] = [2].$$

Two equivalence classes never partially overlap: they are equal or disjoint.

[Added] Worked example: same length, same class

On the set of finite bit strings, define $s \sim t$ when s and t have the same length. Equality of lengths is reflexive, symmetric, and transitive, so this is an equivalence relation. The class of 101 consists of all eight three-bit strings:

$$[101] = \{000, 001, 010, 011, 100, 101, 110, 111\}.$$

This adapts the "equal string length" example in Rosen §9.5, p. 609.

4 Partitions: the geometry of equivalence

Slide sequence: Slides 26–37

Textbook anchor: §9.5, "Equivalence Classes and Partitions," pp. 612–614.

A **partition** of a set A is a collection of subsets, called **parts** or **blocks**, such that:

1. every part is nonempty;
2. the union of all parts is A; and
3. distinct parts are disjoint.

In a picture, imagine the elements of A enclosed in nonoverlapping regions. Every element lies inside one region, and no element lies in two. The takeaway is that a partition records a unique group membership for every element.

4.1 Why classes are equal or disjoint

Suppose $z \in [a] \cap [b]$. Then zRa and zRb. Symmetry turns zRa into aRz, and transitivity with zRb gives aRb. Once the representatives are related, every element related to a is also related to b, so $[a] \subseteq [b]$. The reverse argument gives $[b] \subseteq [a]$. Hence

$$[a] \cap [b] \neq \varnothing \quad \Longrightarrow \quad [a] = [b].$$

Reflexivity places every a in $[a]$, so all classes together cover A. Therefore the equivalence classes of any equivalence relation form a partition.

4.2 Going backward: a partition creates a relation

Given a partition $\mathcal{P}$ of A, define

$$xRy \quad \Longleftrightarrow \quad x \text{ and } y \text{ belong to the same part of } \mathcal{P}.$$

This relation is reflexive and symmetric immediately. For transitivity, if x, y share one part and y, z share another, the two parts overlap at y. Distinct parts cannot overlap, so the parts are the same and x, z share it.

Worked example: list the relation produced by two blocks

Let $A = \{1, 2, 3, 4\}$ and use the partition

$$\{\{1, 3\}, \{2, 4\}\}.$$

All ordered pairs within each block belong to R:

$$R = \{(1, 1), (1, 3), (3, 1), (3, 3), \\ (2, 2), (2, 4), (4, 2), (4, 4)\}.$$

Equivalently,

$$R = (\{1, 3\} \times \{1, 3\}) \cup (\{2, 4\} \times \{2, 4\}).$$

[Added] Worked example: a larger partition, solved systematically

Take the partition

$$\mathcal{P} = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}.$$

The induced equivalence relation is the union of one Cartesian square for each block:

$$R = \{1, 2, 3\}^2 \cup \{4, 5\}^2 \cup \{6\}^2.$$

It therefore has $3^2 + 2^2 + 1^2 = 14$ ordered pairs. This is the efficient way to list or count the pairs without missing reverse pairs or loops. The example is adapted from Rosen §9.5, Examples 12–13, pp. 613–614.

4.3 Grouping subsets by cardinality

Let $A = \{p, q, r\}$ and define a relation on the power set $\mathcal{P}(A)$ by

$$X R Y \iff |X| = |Y|.$$

Equality of cardinalities is reflexive, symmetric, and transitive, so this is an equivalence relation. Its four classes are:

Size	Class
0	$\{\varnothing\}$
1	$\{\{p\}, \{q\}, \{r\}\}$
2	$\{\{p, q\}, \{p, r\}, \{q, r\}\}$
3	$\{\{p, q, r\}\}$

The outer braces matter: a class here is a set whose members are themselves sets. If $|A| = n$, there are $n + 1$ size classes

$$C_k = \{X \subseteq A : |X| = k\}, \qquad k = 0, 1, \ldots, n,$$

and $|C_k| = \binom{n}{k}$.

[Added] Worked example: find the largest size classes when $n = 7$

For a seven-element set, the class sizes are

$$\binom{7}{0}, \binom{7}{1}, \ldots, \binom{7}{7} = 1, 7, 21, 35, 35, 21, 7, 1.$$

The two middle classes, consisting of three-element and four-element subsets, tie for largest at 35. This illustrates the slide's general statement that odd n produces two largest middle classes. See Rosen §6.3, p. 409.

4.4 Counterexamples: similar is not automatically equivalent

On $\mathcal{P}(\{1, 2\})$, let XRY mean $X \cap Y \neq \varnothing$. The relation is symmetric, but it is not reflexive because $\varnothing \cap \varnothing = \varnothing$. It is also not transitive:

$$\{1\} R \{1, 2\}, \qquad \{1, 2\} R \{2\}, \qquad \{1\} \not R \{2\}.$$

One failed property is enough to conclude that a relation is not an equivalence relation.

[Added] Worked example: "within one unit" fails transitivity

On $\mathbb{R}$, define xRy when $|x - y| < 1$. Reflexivity and symmetry hold. However, $2.8 R 1.9$ and $1.9 R 1.1$, while $2.8 \not R 1.1$ because $|2.8 - 1.1| = 1.7$. This Rosen example (Example 7, p. 610) shows a common pattern: local similarity can accumulate until two endpoints are no longer similar.

4.5 Real numbers with the same fractional part

Define aRb on $\mathbb{R}$ when $a - b \in \mathbb{Z}$. Then:

- $a - a = 0 \in \mathbb{Z}$;
- $a - b \in \mathbb{Z}$ implies $b - a = -(a - b) \in \mathbb{Z}$; and
- if $a - b$ and $b - c$ are integers, then their sum $a - c$ is an integer.

Thus R is an equivalence relation, and

$$[a] = \{a + k : k \in \mathbb{Z}\}.$$

For instance,

$$[0.2] = \{\ldots, -1.8, -0.8, 0.2, 1.2, 2.2, \ldots\}.$$

Every class has exactly one representative in $[0, 1)$, namely the fractional part $a - \lfloor a \rfloor$.

4.6 The smallest equivalence relation containing the opening relation

Return one last time to

$$R = \{(a, a), (a, b), (b, a), (b, c)\} \quad \text{on} \quad A = \{a, b, c\}.$$

Because a and b are connected in both directions and b is connected to c, symmetry and transitivity force all three elements into one class. The smallest equivalence relation containing R is therefore

$$A \times A.$$

The added pairs are

$$(b,b),(c,c),(a,c),(c,a),(c,b).$$

The important conceptual shortcut is: treat the original pairs as undirected connections; the connected components become the classes; then include every ordered pair within each component.

5 Optional challenge extensions

Slide sequence: Slides 38–43

These final slides deepen the same ideas. They are not separate topics; each one asks you to reuse a definition more creatively.

5.1 Count reflexive, symmetric relations

Let $|A| = n$. Reflexivity forces all n loops. For each unordered pair of distinct elements $\{a,b\}$, symmetry gives one binary choice: include both (a,b) and (b,a), or include neither. There are $\binom{n}{2}$ such unordered pairs, so the number of reflexive, symmetric relations is

$$2^{\binom{n}{2}}.$$

If the relation must also be antisymmetric, no off-diagonal pair may appear in both directions. Symmetry then forbids every off-diagonal pair, leaving only Δ_A.

5.2 Intersection works; union may fail

If R and S are equivalence relations on A, then $R \cap S$ is an equivalence relation. Both relations contain every loop and every needed reverse. If (a,b) and (b,c) lie in the intersection, they lie in both R and S, so both supply (a,c).

The union can fail transitivity. On $\{1,2,3\}$, take relations induced by the partitions

$$\{\{1,2\},\{3\}\} \quad \text{and} \quad \{\{1\},\{2,3\}\}.$$

Their union contains $(1,2)$ and $(2,3)$ but not $(1,3)$.

5.3 Mutual reachability forms classes

In a directed graph, define $x \sim y$ when x reaches y and y reaches x, now allowing length-zero paths. Reflexivity comes from the length-zero path, symmetry is built into "both directions," and transitivity comes from concatenating paths. The classes are the graph's strongly connected groups.

For edges

$$(1,2),(2,1),(2,3),(3,4),(4,3)$$

on $\{1,2,3,4,5\}$, the classes are

$$\{1,2\}, \qquad \{3,4\}, \qquad \{5\}.$$

The one-way edge $2 \to 3$ cannot merge the first two classes because there is no return path.

5.4 Symmetric and transitive relations on three elements

Suppose R is symmetric and transitive, but not necessarily reflexive, and let

$$B = \{x \in A : xRx\}.$$

If xRy, symmetry gives yRx, and transitivity then forces both xRx and yRy. Thus all pairs of R lie inside $B \times B$, and R is an equivalence relation on B.

For a three-element A, choose B and then choose a partition of B. Sets of sizes $0, 1, 2, 3$ have $1, 1, 2, 5$ partitions, respectively. Therefore the count is

$$\binom{3}{0}(1) + \binom{3}{1}(1) + \binom{3}{2}(2) + \binom{3}{3}(5) = 1 + 3 + 6 + 5 = 15.$$

5.5 Equivalence of subsets via symmetric difference

Let $B \subseteq A$, with $|A| = n$ and $|B| = k$. Define

$$X \sim Y \quad \Longleftrightarrow \quad X \triangle Y \subseteq B,$$

where $X \triangle Y$ contains elements belonging to exactly one of X, Y. The condition says that X and Y may differ inside B, but must agree everywhere outside B. Equivalently,

$$X \setminus B = Y \setminus B.$$

Equality of these outside portions proves reflexivity, symmetry, and transitivity immediately. There are 2^{n-k} ways to fix membership outside B, so there are 2^{n-k} classes. Within a class, the k elements of B are free, giving 2^k subsets per class.

6 Summary

1. A closure keeps every original pair, enforces one property, and is contained in every other relation that does the same job.
2. Reflexive and symmetric closures have direct formulas: $r(R) = R \cup \Delta_A$ and $s(R) = R \cup R^{-1}$.
3. The transitive closure contains exactly the shortcuts for positive-length paths. A cycle, not an empty path, forces a loop.
4. Congruence $a \equiv b \pmod{m}$ means $m \mid (a - b)$, equivalently that a and b have the same remainder modulo m.
5. Addition, subtraction, and multiplication respect congruence. Division requires a multiplicative inverse, which exists exactly when the divisor is relatively prime to the modulus.
6. An equivalence relation is reflexive, symmetric, and transitive. These properties make "belonging to the same group" mathematically stable.
7. Equivalence classes are either equal or disjoint, and together they partition the underlying set.
8. Every partition also defines an equivalence relation: relate elements exactly when they occupy the same part.
9. When repairing a finite relation to an equivalence relation, the connected components of the undirected version become the classes.

7 Key terms glossary

Antisymmetric	If aRb and bRa, then $a = b$.
Closure	The smallest relation containing R that has a specified property.
Congruence modulo m	The relation $a \equiv b \pmod{m}$ defined by $m \mid (a - b)$.
Diagonal relation	$\Delta_A = \{(a, a) : a \in A\}$.
Equivalence class	$[a] = \{x \in A : xRa\}$ for an equivalence relation R.
Equivalence relation	A relation that is reflexive, symmetric, and transitive.
Inverse relation	$R^{-1} = \{(b, a) : (a, b) \in R\}$.
Modulus	The positive integer m that sets the cycle in arithmetic modulo m.
Multiplicative inverse	An integer u satisfying $bu \equiv 1 \pmod{m}$.
Partition	Nonempty, pairwise disjoint subsets whose union is the whole set.
Reflexive	$(a, a) \in R$ for every element of the underlying set.
Representative	Any element used to name its equivalence class.
Symmetric	$(a, b) \in R$ implies $(b, a) \in R$.
Symmetric difference	$X \triangle Y = (X \setminus Y) \cup (Y \setminus X)$.
Transitive	$(a, b), (b, c) \in R$ imply $(a, c) \in R$.
Transitive closure	The relation of all pairs connected by a positive-length path in R.

8 Likely exam questions

1. Compute and justify a transitive closure

Let $R = \{(1, 2), (2, 1), (2, 3)\}$ on $\{1, 2, 3\}$. Find $t(R)$.

Model answer

Reachability from 1 gives $1, 2, 3$; reachability from 2 also gives $1, 2, 3$; from 3 nothing is reachable. Therefore

$$t(R) = R \cup \{(1, 1), (1, 3), (2, 2)\}.$$

The two loops come from the cycle $1 \leftrightarrow 2$, and $(1, 3)$ comes from $1 \to 2 \to 3$.

2. Explain why a claimed closure is not minimal

Why is $A \times A$ usually an inadequate answer for a reflexive closure?

Model answer

It is reflexive and contains R, but it may add off-diagonal pairs not forced by reflexivity. The actual reflexive closure is $R \cup \Delta_A$, which is contained in every reflexive relation containing R.

3. Solve a modular equation and state the legal step

Solve $5x \equiv 8 \pmod{13}$.

Model answer

Because $5 \cdot 8 = 40 \equiv 1 \pmod{13}$, multiply by the inverse 8:

$$x \equiv 8 \cdot 8 = 64 \equiv 12 \pmod{13}.$$

The multiplication is legal because $\gcd(5, 13) = 1$.

4. Prove or refute equivalence

On $\mathbb{Z}$, define aRb when $a - b$ is even. Prove that R is an equivalence relation and describe its classes.

Model answer

$a - a = 0$ is even; the negative of an even integer is even; and the sum of two even integers is even. Thus the relation is reflexive, symmetric, and transitive. There are two classes: the even integers and the odd integers.

5. Move from a partition to a relation

The partition of $A = \{a, b, c, d, e\}$ is $\{\{a, c, e\}, \{b, d\}\}$. How many ordered pairs are in the induced relation?

Model answer

Each block contributes its Cartesian square, so the count is $3^2 + 2^2 = 13$. Every pair of elements from the same block appears, including loops and both directions.

6. Diagnose a false "similarity" relation

On $\mathbb{R}$, let xRy if $|x - y| \leq 2$. Which property fails?

Model answer

Transitivity fails: $0R2$ and $2R4$, but $0 \not R4$ because $|0 - 4| = 4 > 2$. The relation is reflexive and symmetric.

7. Count relations using independent choices

How many reflexive and symmetric relations are there on a four-element set?

Model answer

The four loops are forced. Each of the $\binom{4}{2} = 6$ unordered pairs of distinct elements supplies one independent include-both-or-neither choice. Hence the answer is $2^6 = 64$.

9 Connections

- **Back to earlier relation properties:** closures use the reflexive, symmetric, antisymmetric, and transitive tests developed in Lectures 13–14.
- **Toward partial orders:** Lecture 16 keeps reflexivity and transitivity but replaces symmetry with antisymmetry. That change turns grouping into ordering.
- **Toward graph theory:** transitive closure is reachability, and mutual reachability partitions a directed graph into strongly connected components.
- **Toward counting:** grouping subsets by size introduces $\binom{n}{k}$, while counting symmetric relations introduces independent binary choices.
- **Toward quotient structures:** equivalence classes let later mathematics treat many representatives as a single object, as with residue classes modulo m.